

Amit Kumar

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SUMMARY

M.Sc. Electronics graduate with a strong academic foundation in semiconductor device physics, CMOS fundamentals, and digital/analog IC design principles, seeking to begin an IC Design engineering career at Silicon Labs. Hands-on research experience in thin-film fabrication and precision semiconductor material characterization has built rigorous, process-aware technical thinking directly transferable to IC design and verification workflows. Eager to grow under experienced mentorship in a low-power wireless SoC design environment.

EDUCATION

M.Sc. Electronics

Department of Electronic Science, University of Delhi
South Campus

09/2023 – 07/2025 | New Delhi, India

CGPA: 7.27 / 10

Relevant Coursework: Semiconductor Devices, VLSI Technology, Digital Circuit Design, Analog Circuit Design, Fabrication & Characterization Techniques

B.Sc. (Hons.) Electronic Science

Acharya Narendra Dev College, University of Delhi

11/2020 – 07/2023 | New Delhi, India

CGPA: 7.43 / 10

Key Subjects: Digital Electronics & VHDL, Semiconductor Devices, Microprocessors (8085/8051), Signals & Systems

Class 12

GBSSS Kalkaji, CBSE

04/2019 – 06/2020 | New Delhi, India

Percentage: 81.8%

SKILLS

IC Design Fundamentals

Semiconductor Device Physics (MOSFETs, BJTs, diodes), CMOS circuit fundamentals, digital & analog circuit design, basic timing concepts — coursework-level foundation, actively deepening through self-study and VLSI certification

Digital Design

Digital Circuit Design, Verilog (beginner), basic VLSI concepts — NIELIT VLSI for Beginners certified

Semiconductor Materials & Process

Hands-on experience with thin-film deposition (DC sputtering) and material characterization (XRD, VSM, FMR) — direct exposure to semiconductor process discipline and precision measurement

Simulation & Analysis Tools

Proteus, Multisim, PSpice, MATLAB, Origin Pro — no direct exposure yet to industry IC design EDA tools (Cadence/Synopsys); comfortable and fast at learning new toolchains

Programming

C, C++, Python (basic)

PROFESSIONAL EXPERIENCE

Electronics Intern

07/2023 – 12/2023 | New Delhi, India

Sagedel Tech LLP

Designed and tested analog/digital circuit modules (BJT, 555 Timer IC, Arduino); performed structured functional verification on prototype electronics kits — early exposure to circuit-level design and test discipline.

PROJECTS

Thin Film Fabrication & Semiconductor Characterization

01/2025 – 05/2025

- Fabricated $\text{Fe}_{80}\text{Co}_{20}$ thin films (6–15 nm) via DC sputtering — controlling vacuum ($\sim 10^{-6}$ Torr), Ar pressure (3–5 mTorr), and deposition power (50–200 W) with process-level discipline directly relevant to semiconductor fabrication rigor.
- Applied XRD, VSM, and FMR characterization techniques to extract structural and magnetic parameters, independently learning and operating precision instrumentation with zero prior hands-on experience — demonstrating fast, reliable technical ramp-up ability.
- Interpreted multi-technique measurement data to draw consistent physical conclusions, reflecting the same analytical rigor required in IC design verification and Debug.